

7.3 Halogen Compounds (A Level Only)

Question Paper

Course	CIE A Level Chemistry
Section	7. Organic Chemistry (A Level Only)
Topic	7.3 Halogen Compounds (A Level Only)
Difficulty	Medium

Time allowed: 40

Score: /30

Percentage: /100

Question 1a

This question is about halogenoarene compounds.

State the reagents and name of the mechanism used to convert benzene into chlorobenzene

[2 marks]

Question 1b

Using your answer to (a), write the equation for the generation of the ion which reacts with benzene.

[1 mark]

Question 1c

Draw the mechanism for the chlorination of benzene.

[4 marks]

Question 1d

Explain how the carbon - chlorine bond in chlorobenzene prevents nucleophilic substitution from occurring.

[3 marks]

Question 2a

Predict the products of the following reactions in Fig. 2.1 and draw their structures in the boxes provided. Note that the molecular formula of the final product is given in each case.

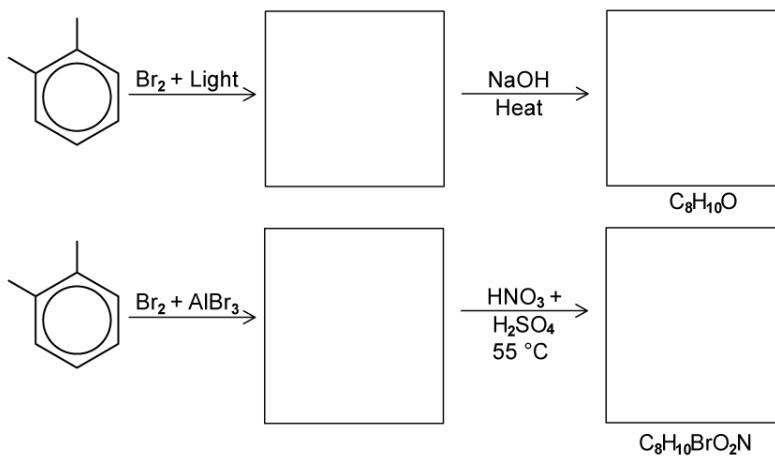


Fig. 2.1

[4 marks]

Question 2b

Write the equation for the generation of the electrophile when methylbenzene reacts with bromine and aluminium bromide.

[1 mark]

Question 2c

Draw the reaction mechanism for the reaction of methylbenzene with bromine and aluminium bromide.

[4]

[3 marks]**Question 2d**

Explain why halogenoalkanes such as bromopropane are more reactive than halogenoarenes such as bromobenzene.

[4 marks]

Question 3a

A student investigated two reactions of phenylethene, $\text{C}_6\text{H}_5\text{CH}=\text{CH}_2$ as shown in Fig. 3.1. First she reacted phenylethene with excess bromine at room temperature to form Compound **A**. She then added aluminium bromide, AlBr_3 to the reaction mixture to form Compound **B**.

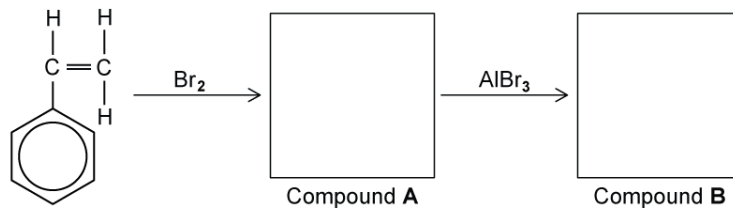


Fig. 3.1

Draw the structures of Compound **A** and **B**.

[2 marks]

Question 3b

Name and outline the mechanism for the formation of Compound **B** from Compound **A**.

[4 marks]

Question 3c

Nitrobenzene was reacted with aluminium chloride and chlorine. Name and draw the structure of the aromatic product formed in this reaction.

[2 marks]

Model Answers: Medium

1a

a) The reagents and the name of the mechanism used to convert benzene into nitrobenzene are:

Reagents:

- AlCl_3
AND
 Cl_2 ; [1 mark]

Mechanism:

- Electrophilic substitution; [1 mark]

[Total: 2 marks]

- You are expected to know the reagents and mechanism for the halogenation of benzene
- **Tip:** Drawing benzene and chlorobenzene can help visualise the mechanism
- One of the hydrogens is replaced / **substituted** by the chloro group
- The benzene ring has a large area of delocalised electrons making it attractive to electrophiles
- Therefore, the mechanism will be electrophilic substitution

1b

The equation for the formation of the ion is:

- $\text{Cl}_2 + \text{AlCl}_3 \rightarrow \text{AlCl}_4^- + \text{Cl}^+$; [1 mark]

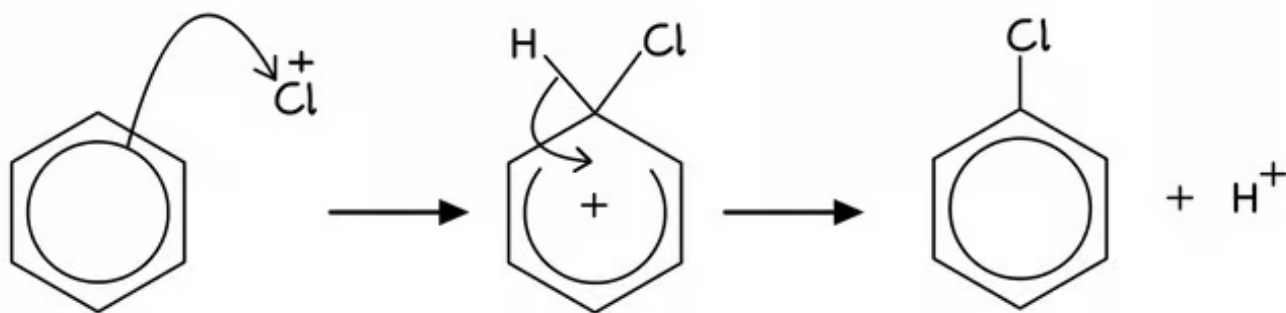
[Total: 1 mark]

- Chlorine gas is bubbled into benzene at room temperature and in the presence of an anhydrous AlCl_3 catalyst to form chlorobenzene
- The AlCl_3 catalyst is also called a halogen carrier and is required to generate the

electrophile (Cl^+)

1c

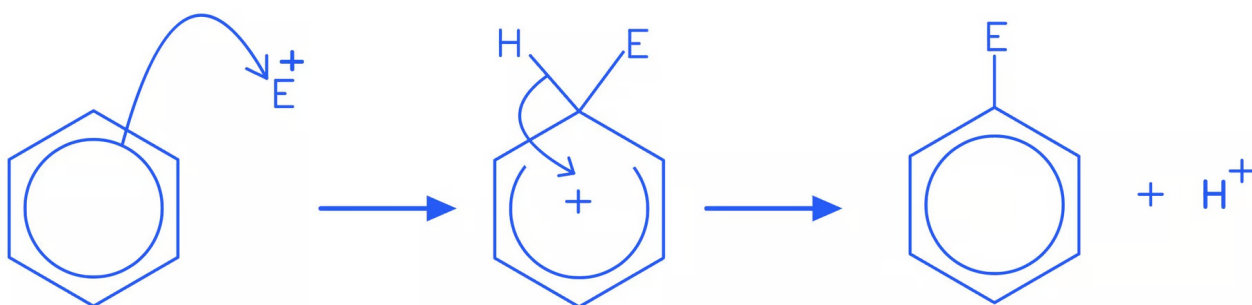
c) The mechanism for the nitration of benzene is:



- Curly arrow going from benzene ring to the Cl^+ ion; [1 mark]
- Correct carbocation structure and positive charge in the (horseshoe) ring; [1 mark]
- Curly arrow going from the middle of the C-H bond towards the positive charge of the benzene cation; [1 mark]
- Correct structure of chlorobenzene

AND H^+ ; [1 mark]**[Total: 4 marks]**

- All electrophilic substitution mechanisms follow the same general pattern:



- The electrophile bonds to the benzene ring, forming an intermediate with a partially delocalised electron system – shown as a horseshoe with + inside

- There are only two locations that you should draw the + charge, either on the electron deficient carbon atom on the benzene ring or in the centre of the benzene ring
- The intermediate then loses a proton, restoring the delocalised electron system

1d

d) The carbon - chlorine bond in chlorobenzene prevents nucleophilic substitution from occurring by:

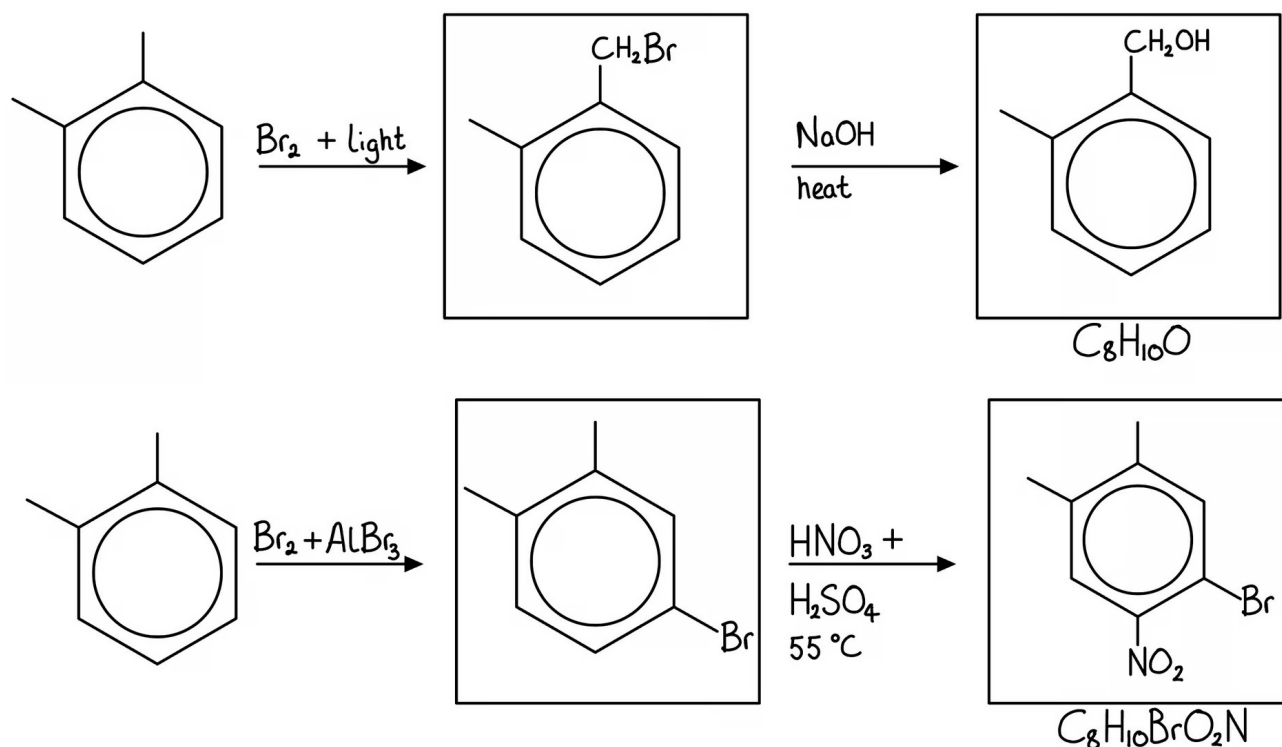
- C-Cl bond is very strong; [1 mark]
- One of the lone pairs of electrons on the chlorine interacts with the π system (of benzene ring); [1 mark]
- C-Cl bond will (therefore) have partial double-bond character; [1 mark]

[Total: 3 marks]

- Halogenoarenes, such as chlorobenzene, do not readily undergo nucleophilic substitution reactions
 - Only under extremely harsh conditions, such as temperatures of 200 °C and a pressure of 200 atmospheres, will the chlorine in chlorobenzene get replaced by a nucleophile such as a hydroxide (OH^-) ion
- The partial double bond of the C-Cl will strengthen it
- The question asks specifically about the C-Cl bond in chlorobenzene
 - Nucleophiles will be repelled by the delocalisation in the benzene ring, but this is not what the question is asking about!
- Halogenoalkanes such as chloroethane can take part in nucleophilic substitution reactions
 - A nucleophile, such as a hydroxide (OH^-) ion, will attack the slightly positive carbon atom in the carbon chain
 - A covalent bond is formed between that carbon atom and the nucleophile which causes the carbon-halogen bond to break

2a

a) The correct products are:



- Each correct structure scores [1 mark]

[Total: 4 marks]

- The first reaction sequence involves substitution of a hydrogen atom with a bromine atom on the methyl group
 - These reagents will allow free radical substitution to occur, but not electrophilic substitution
 - Nucleophilic substitution can then occur to form the alcohol
- The second reaction sequence involves a halogen carrier so electrophilic substitution can occur
 - Another suitable halogen carrier could be FeBr_3
 - You are allowed any orientation of the bromine atom due to two methyl groups present

groups present

- This is also the same of the next compound
- Concentrated HNO_3 and concentration H_2SO_4 will react to form the NO_2^+ electrophile
- This undergoes electrophilic substitution with the benzene ring

2b

b) The equation for the generation of the electrophile is:

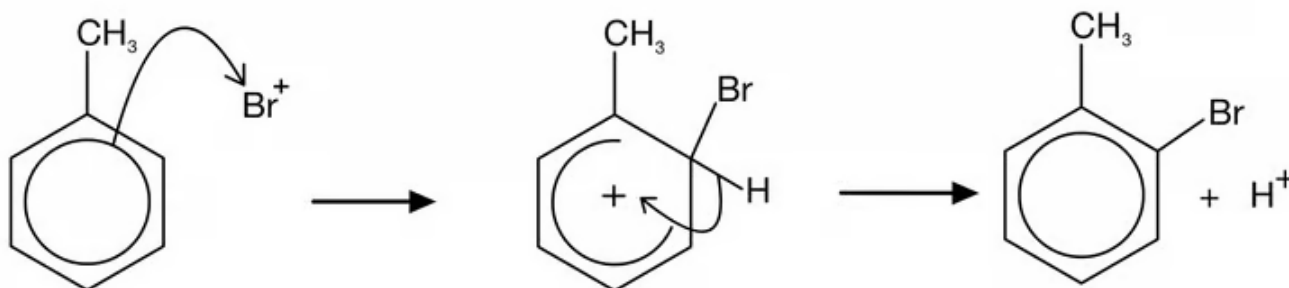
- $\text{AlBr}_3 + \text{Br}_2 \rightarrow [\text{AlBr}_4]^- + \text{Br}^+$; [1 mark]

[Total: 1 mark]

- The AlBr_3 catalyst is called a halogen carrier and is required to generate the electrophile Br^+

2c

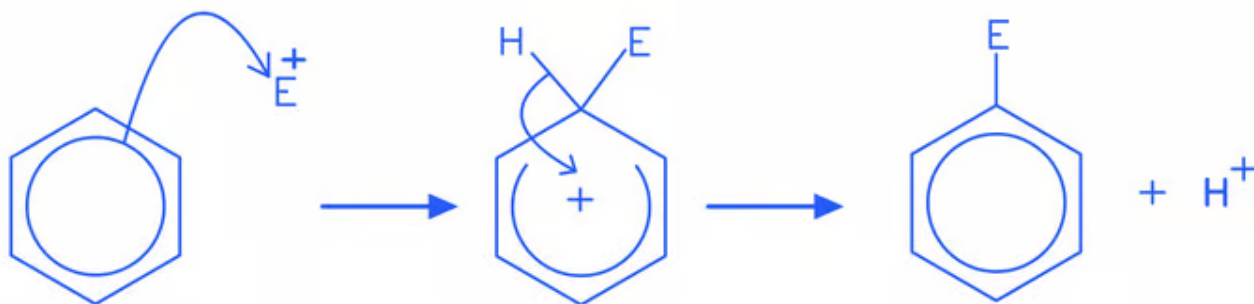
c) The mechanism is:



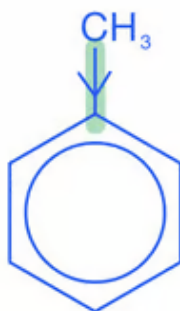
- Curly arrow going from benzene ring to the Br^+ ion; [1 mark]
- Correct carbocation structure and positive charge in the (horseshoe) ring; [1 mark]
- Curly arrow going from the middle of the C-H bond towards the positive charge of the methylbenzene cation; [1 mark]
- Correct structure of 2-bromomethylbenzene or 4-bromomethylbenzene
AND
 H^+ ; [1 mark]

[Total: 4 marks]

- The general mechanism for electrophilic substitution is always the same so make sure you learn it thoroughly



- The methyl group is electron donating, so it is 2,4- directing, therefore your bromomethylbenzene product must be either 2-bromomethylbenzene or 4-bromomethylbenzene



2d

d) Halogenoalkanes such as bromopropane are more reactive than halogenoarenes such as bromobenzene because:

- Halogenoalkanes / bromopropane contain a polar C-Br bond / δ^+ C atom and δ^- Br atom; [1 mark]
- Nucleophiles can attack δ^+ C atom; [1 mark]
- C-Br bond in bromobenzene is very strong; [1 mark]
- Lone pairs of electrons on the bromine interact with the π system (of benzene ring); [1 mark]

[Total: 4 marks]

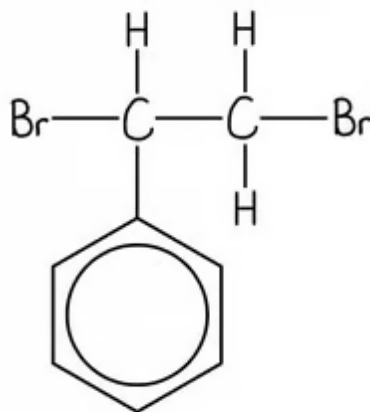
- Halogenoarenes are very unreactive compared to halogenoalkanes
- The difference in reactivity between the two compounds is because of the

carbon-halogen bond strengths

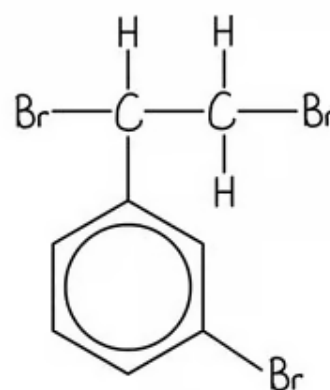
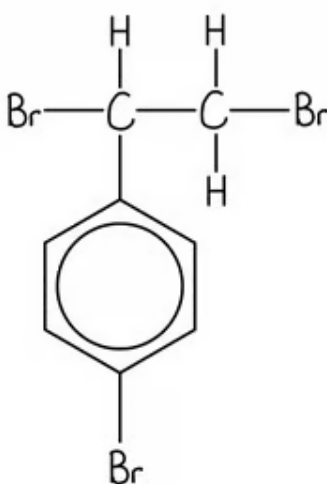
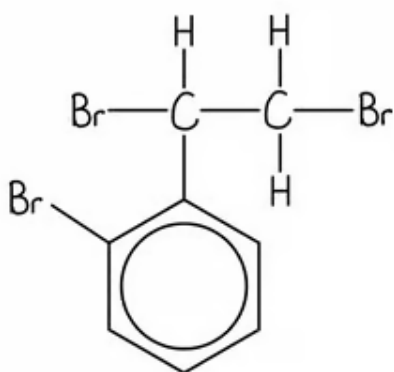
3a

a) The structures of Compounds **A** and **B** are:

- Compound **A**



- Correct structure; [1 mark]
- Compound **B**:

Any **one** from the following:

; [1 mark]

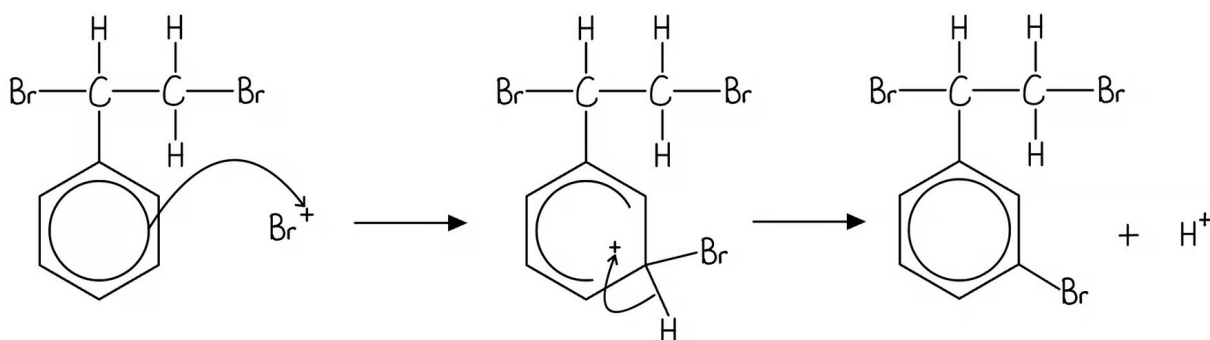
[Total: 2 marks]

- **Remember:** The halogenation of a benzene ring requires a halogen carrier, in this case, aluminium bromide, AlBr_3
- An alkene will undergo electrophilic addition with a halogen molecule which will occur at room temperature
- There are three possible isomers for Compound **B** as the bromine atom can be substituted onto either the 2nd, 3rd or 4th carbon atom

3b

b) The name and outline of the mechanism to form Compound **B** from Compound **A** is:

- Electrophilic substitution; [1 mark]
- The mechanism is:



- Curly arrow from the ring inside benzene to the Br^+ electrophile; [1 mark]
- Correct intermediate structure; [1 mark]

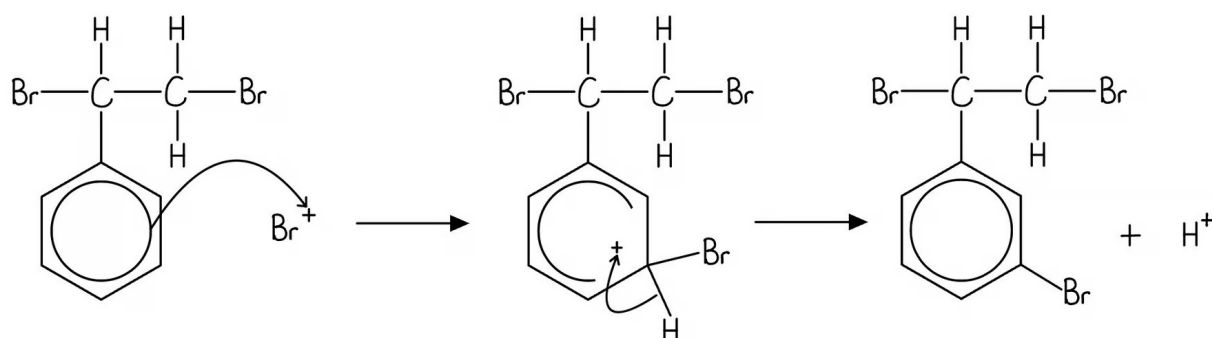
*The horseshoe must cover at least half of the ring with the 'gap' at the correct C
It must cover more than half of the benzene but must not extend past carbon 2 or 4*

- There are three possible isomers for Compound **B** as the bromine atom can be substituted onto either the 2nd, 3rd or 4th carbon atom

3b

b) The name and outline of the mechanism to form Compound **B** from Compound **A** is:

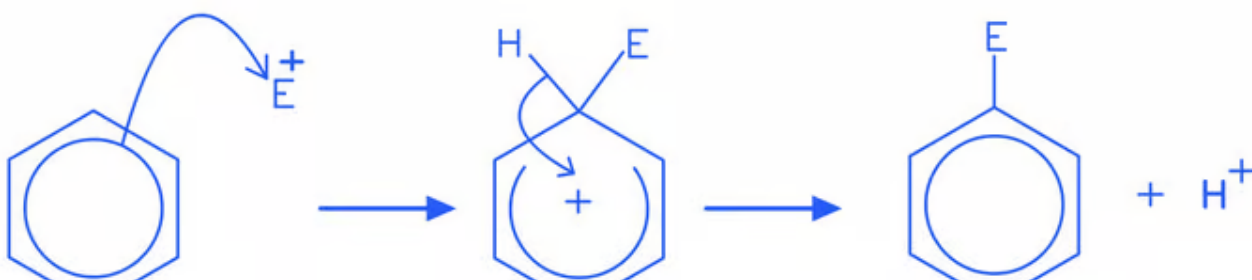
- Electrophilic substitution; [1 mark]
- The mechanism is:



- Curly arrow from the ring inside benzene to the Br^+ electrophile; [1 mark]
- Correct intermediate structure; [1 mark]
The horseshoe must cover at least half of the ring with the 'gap' at the correct C. It must cover more than half of the benzene but must not extend past carbon 2 or 4.
- Curly arrow from the C-H bond back to the + within the ring
AND
Correct products shown; [1 mark]

[Total: 4 marks]

- The general mechanism for electrophilic substitution:

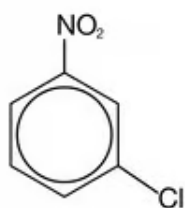


- The horseshoe in the benzene ring intermediate must cover at least half of the ring with the 'gap' at the carbon being substituted
 - It must cover more than half of the benzene but must not extend past carbon 2 or 4

3c

c) The name and product formed is:

- 3-chloronitrobenzene
OR
5-chloronitrobenzene; [1 mark]



; [1 mark]

[Total: 2 marks]

- The nitro group (NO_2) is an electron-withdrawing group, so it is 3 and 5 directing
- Therefore the chlorine atom will be substituted onto either the 3rd or 5th carbon

